



External Beam Rotation II: Advanced Treatment Planning

Rotation mentors:

Knowledge:.....Practicum.....

Rotation Dates:

Reviews:

- Bi-Weekly Progress Review by the Mentor.
- End of rotation review by the Residency Director.

This assignment sheet outlines the requirements for successful completion of the External Beam Rotation II. The resident must complete the knowledge and practical factor items listed below. Satisfactory performance is documented by staff signoff with dates and passing of the end of rotation oral exam.

Key:

PD=Residency Program Director

APD=Associate Program Director

RM=Rotation Mentor

SD=Staff Dosimetrist

Essential Reading

- ICRU Report 50 and 62.
- Penn departmental practice guidelines related to IMRT planning
- Treatment Planning in Radiation Oncology, 4th Edition, FM Khan, JP Gibbons, and PW Spertudo, Wolters Kluwer, 2016
 - Chapters 9, 12, 20, 22
- TG 50 Basic applications of multileaf collimators
- TG 65 Tissue inhomogeneity corrections for megavoltage photon beams
- TG 119 IMRT commissioning: Multiple institution planning and dosimetry comparisons
- TG 120 Dosimetry tools and techniques for IMRT
- TG 142 Quality assurance of medical accelerators
- TG 166 The use and QA of biologically related models for treatment planning
- TG 263 Standardizing nomenclatures in radiation oncology
- RTOG 0631 Spine metastasis
- RTOG 0813 Lung SBRT
- RTOG 0915 Lung SBRT
- Eclipse Photon and Electron Algorithms Reference Guide (myVarian)
- Eclipse Photon and Electron Instructions for Use (myVarian)
- IMRT Prostate Boost (Phase 2) Using Base Dose Plan (myVarian)

Supplemental Reading

- The Modern Technology of Radiation Oncology, Volume 2, Editor J. Van Dyk, Medical Physics Publishing, Copyright 2005.
- Chapters 4, 5, 6
- The Physics of Radiation Therapy, 5th edition, FM Khan and JP Gibbons, Lippincott Williams & Wilkins, 2014
- Chapters 20, 21, 22, 26
- Image-Guided IMRT, Editors T. Bortfeld, R. Schmidt-Ullrich, W. de Neve, and D. Wazner, Springer-Verlag, Heidelberg, German, 2006.
- Intensity-Modulated Radiation Therapy: The State of the Art, Editors J. Palta and T. Mackie, Medical Physics Publishing Corporation, 2003.
- A Practical Guide to Intensity-Modulated Radiation Therapy, Memorial Sloan-Kettering Cancer Center, Medical Physics Publishing Corporation, 2003
- Yu, C. X. et al., "Planning and delivery of intensity-modulated radiation therapy" *Med. Phys.* **35** (12) 5233-5241 (2008).
- Galvin, J. M. et al. "Implementing IMRT in clinical practice: a joint document of the ASTRO and the AAPM" *Int J Radiat Oncol Biol Phys.* **58** (5), 1616-1634 (2004).
- Ezzell, G.A. et al. "Guidance document on delivery, treatment planning, and clinical implementation of IMRT: Report of the IMRT subcommittee of the AAPM radiation therapy committee", *Med. Phys.* **30** (8), 2089 – 2115 (2003).
- Otto, K. "Volumetric modulated arc therapy: IMRT in a single gantry arc" *Med. Phys.* **35**(1), 310-317 (2008)

- Binny, D. *et al.* "Photon optimizer (PO) vs progressive resolution optimizer (PRO): a conformality- and complexity-based comparison for intensity-modulated arc therapy plans" *Medical Dosimetry* **43**, 267-275 (2018)
- Van Dyk, J., *et al.* "Commissioning and quality assurance of treatment planning computers." *International Journal of Radiation Oncology Biology Physics* **26**(2) (1993): 261-273.
- Xia, P., Verhey, L.J. "Delivery systems of intensity-modulated radiotherapy using conventional multileaf collimators" *Medical Dosimetry* **26**(2): 169-177 (2001)
- Mackie, T.R. "History of tomotherapy" *Phys. Med. Biol.* **51**, R427-R453 (2006)
- Pinnacle³ White Paper – Collapsed Cone Convolution Superposition and Delta Pixel Beam
- Pinnacle³ White Paper – Direct machine parameter optimization (DMPO)

Task Group Reports

1. TG 50 Basic applications of multileaf collimators
2. TG 65 Tissue inhomogeneity corrections for megavoltage photon beams
3. TG 119 IMRT commissioning: Multiple institution planning and dosimetry comparisons
4. TG 120 Dosimetry tools and techniques for IMRT.
5. TG 142 Quality assurance of medical accelerators
6. TG 166 The use and QA of biologically related models for treatment planning
7. TG 263 Standardizing nomenclatures in radiation oncology

RTOG Reports

1. RTOG 0631 Spine metastasis
2. RTOG 0813 Lung SBRT
3. RTOG 0915 Lung SBRT

Eclipse Reference materials from myvarian.com:

1. IMRT/RapidArc H&N Procedure
2. IMRT Prostrate procedure
3. IMRT Prostrate boost with Base dose plan
4. RapidArc overview, workflow and QA procedures
5. Eclipse Algorithm Reference Guides V13 and V15.5
6. RapidPlan Knowledge-Based Planning Bibliography

Review Articles

1. Yu, C. X. *et al.*, "Planning and delivery of intensity-modulated radiation therapy" *Med. Phys.* **35** (12) 5233-5241 (2008).
2. Galvin, J. M. *et al.* "Implementing IMRT in clinical practice: a joint document of the ASTRO and the AAPM" *Int J Radiat Oncol Biol Phys.* **58** (5), 1616-1634 (2004).
3. Ezzell, G.A. *et al.* "Guidance document on delivery, treatment planning, and clinical implementation of IMRT: Report of the IMRT subcommittee of the AAPM radiation therapy committee", *Med. Phys.* **30** (8), 2089 – 2115 (2003).

4. IMRT collaborative working group "IMRT: Current status and issues of interest" *Int. J. Rad. Oncol. Biol. Phys.* **51** (4), 880 – 914(2001).
5. Otto, K. "Volumetric modulated arc therapy: IMRT in a single gantry arc" *Med. Phys.* **35**(1), 310-317 (2008)
6. Binny, D. et al. "Photon optimizer (PO) vs progressive resolution optimizer (PRO): a conformality- and complexity-based comparison for intensity-modulated arc therapy plans" *Medical Dosimetry* **43**, 267-275 (2018)
7. Van Dyk, J., et al. "Commissioning and quality assurance of treatment planning computers." *International Journal of Radiation Oncology Biology Physics* **26**(2) (1993): 261-273.
8. Kim, S., Zhu, T.C., Palta, J.R. "An equivalent square field formula for determining head scatter factors of rectangular fields" *Med. Phys.* **24**(11), 1770-1774
9. Xia, P., Verhey, L.J. "Delivery systems of intensity-modulated radiotherapy using conventional multileaf collimators" *Medical Dosimetry* **26**(2): 169-177 (2001)
10. Mackie, T.R. "History of tomotherapy" *Phys. Med. Biol.* **51**, R427-R453 (2006)
11. Pinnacle³ White Paper – Collapsed Cone Convolution Superposition and Delta Pixel Beam
12. Pinnacle³ White Paper – Direct machine parameter optimization (DMPO)

Discussion Points

1. Discuss the differences between conventional, 3D-conformal, and IMRT/VMAT treatment and treatment planning. The resident should understand the fundamental differences and be able to discuss the pros and cons of the different methods including the use of objective functions.
2. Discuss the different types of IMRT planning techniques, including forward planned segments, inverse beamlet based optimization, inverse segment optimization, tomotherapy, intensity modulated arc therapy, etc.
3. Discuss the role of contouring in IMRT planning and how the requirements may differ from 3D CRT.
4. Discuss the different types of optimization algorithms and demonstrate understanding of when to apply them.
5. Discuss the differences between dynamic multileaf collimator (DMLC) and segmental multileaf collimator (SMLC) leaf sequencing algorithms in terms of delivery parameters and dose distributions

Training/Observation

1. The mentor(s) should train the resident on the process of IMRT treatment planning in Eclipse Varian TPS. The training should include a step-by-step demonstration of the planning process from contouring to evaluation.
2. The resident should observe a dosimetrist planning clinical IMRT and RapidArc cases as indicated below for the initial three weeks of the rotation.
3. As needed, the physicist mentors will be available for additional training and discussion.
4. The mentor SP or SD will sign off on each resident-planned case.

Clinical Cases to plan: 2 cases per week independently, but under supervision of mentors after 2-3 weeks of observation, lectures and library cases as cleared by the mentors.

Rotation Competency Sign-Off

Knowledge Factors

Discuss the differences between conventional, 3D-conformal, and IMRT treatment and treatment planning. The resident should understand the fundamental differences and be able to discuss the pros and cons of the different methods including the use of objective functions.

(RM) Signature / Date

Discuss the different types of IMRT planning techniques, including forward planned segments, inverse beamlet based optimization, inverse segment optimization, tomotherapy, intensity modulated arc therapy, etc...

(RM) Signature / Date

Discuss the role of contouring in IMRT planning and how the requirements may differ from 3D CRT.

(RM) Signature / Date

Discuss the different types of optimization algorithms and demonstrate understanding of when to apply them.

(RM) Signature / Date

Discuss how planning methodology may be different between VMAT and IMRT.
(RM) Signature / Date

Discuss the contouring and planning process for a SBRT lung case.

(RM) Signature / Date

Discuss the differences between dynamic multileaf collimator (DMLC) and segmental multileaf

collimator (SMLC) leaf sequencing algorithms in terms of delivery parameters and dose distributions(RM) <i>Signature / Date</i>	
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Discuss the different QA procedures (Mapcheck, Arc Check, Portal dosimetry and Film QA).

(RM) <i>Signature / Date</i>	
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Discuss the different QA process for IMRT/ Rapid Arc treatment plans.

(RM) <i>Signature / Date</i>	
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Practical Factors

A. Training & Observation

Attendance of training provided by SD or designee on the process of IMRT /RA treatment planning – including a step-by-step demonstration of the planning process from contouring to evaluation.

(SD) <i>Signature / Date</i>	
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Observe a SD planning a clinical IMRT treatment for each of the following sites:

Prostate

(SD) <i>Signature / Date</i>	
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Whole Pelvis

(SD) <i>Signature / Date</i>	
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Brain

(SD) <i>Signature / Date</i>	
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Head/Neck

(SD) <i>Signature / Date</i>	
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Lung

(SD) <i>Signature / Date</i>	
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Other Clinical Site:

(SD) <i>Signature / Date</i>	
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B. Planning Techniques

Residents should maintain disease-site-specific notes of their library and clinical case planning. The log should include an anonymized history & present illness, site, planning strategies, how the patient is simulated, and how the target is localized.

Plan an EZFluence case (library case) after dosimetry demonstration. The rotation mentor will review, discuss, and make sure the plan quality is acceptable.

(RM) Signature / Date	
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Plan a Prostate IMRT case with 5, 7, and 9 IMRT beams. A 1 and 2 arcs treatment should be planned as well using RapidArc(RA). The resident should present their results and their conclusions on the effect of the number of beams on plan quality to the faculty mentor.

(RM) Signature / Date	
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Plan a brain or sinus IMRT case with planar beams only and then non-coplanar beams. Discuss the advantages and disadvantages of non-coplanar beams in IMRT with your faculty mentor. A 1 and 2 arcs treatment should be planned as well using RapidArc(RA).

(RM) Signature / Date	
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Contour and plan a SBRT Lung case (library cases only) after dosimetry demonstration. The rotation mentor will review the plan and discuss the advantages of 3D versus IMRT versus RapidArc planning for SBRT.

(RM) Signature / Date	
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C. Clinical Cases

With clearance from the mentor, plan IMRT or VMAT cases for the clinic for the sites listed below. Each clinical case will be preceded by a practice case of the same site, approved by the mentor, for a total of 14 cases. A case is defined as a new plan on a different CT dataset. Various beam arrangements, plan iterations, or conedown planned on the same CT dataset are considered 1 case.

The mentor (RM) will sign off on each resident-planned case. If no clinical cases are available during the rotation, library cases may be used.

Prostate

(RM) Signature / Date

Whole Pelvis

(RM) Signature / Date

Brain

(RM) Signature / Date

Head/Neck

(RM) Signature / Date

Lung

(RM) Signature / Date

GYN:

(RM) Signature / Date

Breast / APBI:

(RM) Signature / Date

Residency Director (PD/APD)
Signature / Date